CSE4007
Artificial Intelligence

Lecture 0: Overview

Department of Data Science
Hanyang University

Class Logistics

- Time & Place
  - Wed. 9:00am – 12:00pm
  - FTC Intelligence Computing Room 4
- Instructor: Prof. Sungyong Baik
  - dsybaik@hanyang.ac.kr
- Teaching Assistant: Heesun Jung
  - jheesun@hanyang.ac.kr
  - Responsible for marking assignments, exams
- Course materials
  - https://lms.hanyang.ac.kr
  - Lectures, other course materials, and announcements will be posted

References

- Deep Learning
  - By Goodfellow, Bengio, and Courville
  - Freely available at
    https://www.deeplearningbook.org/
- Pattern Recognition and Machine Learning
  - By Bishop
- Convolutional Neural Networks for Visual Recognition
  - Stanford CS231n
  - By Li, Johnson, and Yeung
- Recent research papers

Grading

- Attendance: 5%
- 3 (programming) assignments/Project proposal: 30%
- Midterm: 30%
- Final Exam: 35%

- Late Submission Policy
  - 25% off per late day

- No cheating, please!
  - You can have discussions, but do not copy others' works
  - Any form of plagiarism, cheating, academic misconduct will lead to **grade F** on this course (**and even other courses** in the same semester) and other disciplinary actions (including **degree cancellation**).
    - **You won't be able to retake the course, if you get caught for plagiarism**
  - Copying work from any AI system (e.g., ChatGPT, Gemini) is prohibited.

Tentative Schedule

- Computer Vision Applications
  - Object Detection
  - Object Segmentation
- Sequence Modeling
  - RNN
  - LSTM
  - Image Captioning
- Attention & Transformers
  - Language Models
- Generative Models Part 1
  - Generative Adversarial Networks (GANs)
- Generative Models Part 2
  - Variational Autoencoder (VAE)
- Generative Models Part 3
  - Diffusion Model

Tentative Schedule

- Midterm
  - The week of April 14~18
    - Most likely, April 16
- Assignment 1
  - Comes out on Mar 24th
  - Due on April 6th
- Assignment 2
  - Comes out on April 21st
  - Due on May 4th
- Assignment 3
  - Comes out on May 19th
  - Due on June 1st
- Final Exam
  - The week of June 9~13
    - Most likely, June 11th